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Software Engineering graduate focused on data science, analytics, and applied AI. Experienced building end-to-end scraping and market-intelligence pipelines, NLP clustering with SentenceTransformers, computer vision landmark extraction, dashboards, and Python systems that connect data collection to business decisions.

Education

Arab Academy for Science, Technology and Maritime Transport

B.Sc. in Software Engineering

Cairo

Feb 2026

Oxford Summer Course

Java Programming & Problem Solving

Oxford, UK

July 2018

Experience

Happy Vision

Data Analyst Intern

Sheikh Zayed - Giza

Jan 2026 - Present

- Automated Large-Scale Data Extraction: Engineered and deployed web scraping scripts to extract comprehensive data for over 10,000 products, ensuring data integrity and handling site-specific bottlenecks.
- Multilayered Product Classification: Developed a classification system to organize raw scraped data into distinct segments (e.g., Kids, Adults), streamlining the cataloging process for downstream analysis.
- NLP-Driven Product Clustering: Leveraged the all-MiniLM-L6-v2 SentenceTransformer model to calculate cosine similarity across product descriptions, successfully identifying and clustering near-duplicate items within categories to improve database efficiency.

X-inity

AI Intern

Maadi, Cairo

August - September 2025

- Gained hands-on experience with MediaPipe for pose/landmark detection, adapting it for domain-specific applications
- Developed a program to automatically extract body landmarks from Karate Katas (moves) using MediaPipe's pose estimation models.
- Strengthened skills in Python, OpenCV, computer vision pipelines, data preprocessing, and ML model integration.

Projects

AI-Powered - Scraping, Categorization and Market Intelligence | <https://github.com/mady-mohamed/WebScraping>

- Built a Streamlit application that scrapes Shopify product data, cleans and normalizes records, categorizes products, clusters similar items, tracks price changes, and supports market research.
- Implemented concurrent scraping, background jobs, semantic search, product comparison dashboards, t-SNE market white-space visualization, cluster summaries, CSV export, and a local LM Studio-powered market advisor.
- Designed a modular pipeline using Python, Pandas, Requests, aiohttp, BeautifulSoup4, SentenceTransformers, PyTorch, NumPy, NetworkX, scikit-learn, SQLite, and OpenAI-compatible chat completions

Python Pacman Game with AI Implementation | <https://github.com/mady-mohamed/Pacman>

Engineered a complete Pacman game from scratch in Python, demonstrating a strong command of game development, object-oriented programming, and AI algorithms.

- Designed intelligent ghost AI using A* search for efficient pathfinding (predator mode) and a Greedy algorithm for evasive maneuvers (prey mode).

- Enhanced application performance by over 20% by identifying bottlenecks in the A* algorithm, re-implementing those critical sections in C++, and integrating them with Python using the ctypes library.
- Implemented core gameplay mechanics including collision detection, player/ghost movement, scoring, and level progression using Pygame.
- Developed a modular and well-organized codebase for clear handling of game logic, settings, and UI components.

Skills & Interests

Design & Development Tools

Figma (UI/UX Design, Prototyping), Git & GitHub, Google Looker Studio

Language

English (Fluent), Arabic (Native)

Machine Learning & Data

Python, SQL, Pandas, NumPy, scikit-learn, PyTorch, NLP, SentenceTransformers, cosine similarity, clustering, t-SNE, statistical modeling

Computer Vision

MediaPipe, OpenCV, pose/landmark detection, data preprocessing, ML model integration

Tools & Visualization

Google Looker Studio, Streamlit, SQLite, Git/GitHub, Figma, CSV export, semantic search workflows

Software Engineering

Java, JavaScript, HTML/CSS, C++ optimization, OOP, web scraping, BeautifulSoup4, aiohttp, Requests, modular architecture